

MOONLIGHT EVENTS

PROJECT DOCUMENTATION

Submitted By

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1. PROBLEM DEFINITION

Background

Many cultural festivals around the world are poorly represented online, especially local festivals that showcase traditions, talents, creativity, music, dance, and cultural heritage. Young people and visitors often struggle to discover cultural events, learn about traditions, or explore creative talents connected to these festivals.

Most existing festival platforms focus only on event promotion and do not adequately support cultural education, talent discovery, or artistic creativity.

Problem Statement

The project aims to solve the problem of limited access to information about local and international festivals while also promoting creative talents associated with these cultural events.

The website provides:

- Information about festivals around the world
- Cultural education and awareness
- Festival image galleries
- Talent discovery opportunities
- Support for creative expression and cultural exchange

- Easy navigation and festival search functionality

Main Goal

To spread awareness about global and local festivals while discovering and promoting creative talents through a modern web platform.

Objectives

1. To provide users with information about cultural festivals.
 2. To create awareness of cultural traditions and celebrations.
 3. To support talent discovery and creativity.
 4. To allow users to explore festivals through images and descriptions.
 5. To provide an interactive and responsive user experience.
 6. To encourage cultural exchange among users.
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2. DESIGN SPECIFICATIONS

Software Requirements

Software	Purpose
React.js	Frontend framework
Vite	Development environment
Bootstrap 5	Responsive styling
React Router DOM	Routing/navigation
AOS (Animate On Scroll)	Scroll animations
jsPDF	PDF generation
Git & GitHub	Version control
Vercel	Deployment

Hardware Requirements

Hardware	Minimum Requirement
RAM	4GB
Processor	Intel Core i3 or equivalent
Storage	1GB free space

Hardware	Minimum Requirement
Internet Connection	Required

Functional Requirements

The system should:

- Display festival information
- Allow users to search for festivals
- Display festival images in a gallery
- Navigate between pages
- Show featured talents
- Allow users to explore cultural information
- Display responsive layouts on mobile and desktop
- Provide smooth scrolling experience
- Display visitor count

Non-Functional Requirements

The system should:

- Be responsive
- Be visually attractive
- Be easy to navigate
- Load quickly
- Support modern browsers
- Provide smooth animations

Color Scheme

Color	Usage
#0B0F1A	Navbar background
#c60707	Accent red
#f7f8fc	Light section background
White	Text and cards

Typography

Font	Usage
Poppins	Body text

Font	Usage
Montserrat	Headings

3. SYSTEM FEATURES

Homepage

The homepage contains:

- Hero section
- Festival listings
- Mini About section
- Featured talents section
- Gallery preview
- Footer

Search System

Users can search for festivals using the navbar search feature.

Features include:

- Smooth scrolling
- Search filtering
- "No items found" display
- Auto-clearing search input after submission

Festival Gallery

Displays festival images in a responsive grid layout.

Featured Talents

Displays talented individuals connected to cultural creativity.

Visitor Counter

Tracks the number of visits using:

- localStorage
- sessionStorage

Routing System

The application uses React Router for:

- Home page
 - About page
 - Contact page
 - FAQ page
 - Gallery page
-

4. FLOWCHARTS

Festival Search Flowchart

```
START
|
User enters search text
|
Clicks search button
|
Check if input is empty
|----- YES -----> Stay on page
|
NO
|
Store submitted search
|
Scroll to festival section
|
Filter festival list
|
Display results
|
END
```

Visitor Counter Flowchart

```
START
|
Component loads
|
Check sessionStorage
```

```

|
Has current session visited?
|----- YES -----> Display existing count
|
NO
|
Get count from localStorage
|
Increment count
|
Save updated count
|
Set sessionStorage visited = true
|
Display count
|
END

```

Navigation Flowchart

```

START
|
User clicks navigation link
|
React Router changes route
|
Page component loads
|
ScrollToTop component activates
|
Page scrolls to top
|
Display selected page
|
END

```

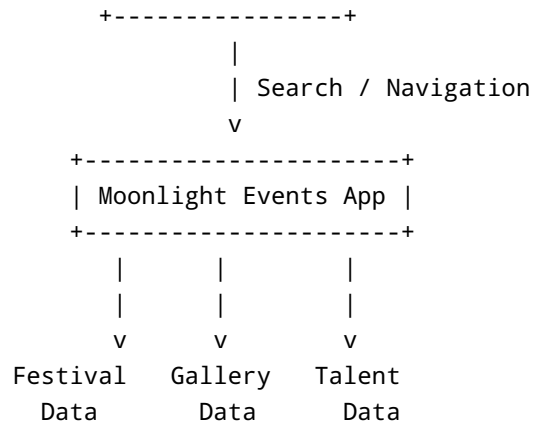
5. DATA FLOW DIAGRAM (DFD)

Level 0 DFD

```

+-----+
|      USER      |
+-----+

```



Level 1 DFD

```

USER
|
| enters search
v
Search Component
|
| sends search term
v
Festival Filter Logic
|
| filters festival data
v
Festival List
|
| displays results
v
USER
  
```

6. DATABASE / DATA STRUCTURE

Festival Data Structure

```

{
  "id": 1,
  "name": "Diwali",
  "country": "India",
  "religion": "Hinduism",
}
  
```

```
"month": "November",  
"description": "Festival of lights",  
"image": "image-url"  
}
```

Talent Data Structure

```
{  
  "id": 1,  
  "name": "Amina Yusuf",  
  "country": "Nigeria",  
  "talent": "Cultural Dancer",  
  "description": "Preserving traditions through dance",  
  "image": "image-url"  
}
```

7. IMPLEMENTATION DETAILS

Frontend

The frontend was built using React.js and Bootstrap.

State Management

React hooks used include:

- useState
- useEffect
- useLocation

Animations

AOS library was used for:

- Fade animations
- Zoom animations
- Flip animations
- Scroll-triggered effects

Routing

React Router DOM was used for page navigation.

Deployment

The project was deployed using Vercel and connected to GitHub for automatic deployment updates.

8. TESTING

The application was tested for:

- Mobile responsiveness
- Navigation
- Search functionality
- Smooth scrolling
- Routing
- Gallery rendering
- Visitor counter functionality

Browsers tested:

- Google Chrome
 - Microsoft Edge
 - Mobile browsers
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9. CHALLENGES ENCOUNTERED

1. Search scrolling behavior
 2. Mobile responsiveness issues
 3. Navbar active link handling
 4. Hero image display on deployment
 5. React Strict Mode affecting visitor count
 6. Route scroll positioning
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10. SOLUTIONS IMPLEMENTED

Problem	Solution
Double visitor counting	Used sessionStorage
Route opening at bottom	ScrollToTop component
Mobile overflow	Responsive Bootstrap classes
Search visibility	Smooth scroll implementation

Problem	Solution
Deployment image issue	Replaced unsupported image URL

11. CONCLUSION

The Moonlight Events web application successfully provides users with a platform to explore festivals, discover creative talents, and learn about global cultures.

The system combines modern frontend technologies, responsive design, smooth animations, and interactive features to create an engaging user experience.

The project also promotes cultural exchange and supports creative expression through its featured talents section.

12. FUTURE IMPROVEMENTS

Future improvements may include:

- Backend database integration
 - User authentication
 - Real talent submissions
 - Festival ticket booking
 - Live event updates
 - Comment and review system
 - Admin dashboard
 - Dark/light mode toggle
 - AI-based festival recommendations
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13. REFERENCES

1. React.js Documentation
2. Bootstrap Documentation
3. React Router Documentation
4. AOS Documentation
5. jsPDF Documentation
6. Vercel Documentation
7. GitHub Documentation